

DATA SCIENCE

Overview

Data Science is a rapidly growing academic discipline fueled by the proliferation of rich and complex data emerging from activities in science, industry, and governments. As a result, there is strong demand for data science professionals today in Iowa as well as across the nation and globe, and this market is expected to continue to grow in the next decade. The data science programs are intended for students who wish to study the data science discipline for its own sake as well as for students studying any discipline at Iowa State University with the goal of enabling them to work in data science. The courses in the data science program are designed to provide students with the requisite background that would enable them to take jobs with significant data science components, e.g., establishing and operating data analysis pipelines. The capstone will provide an opportunity for students to apply data science concepts to a domain problem while working in a multi-disciplinary team setting.

The Data Science major is intended for students with strong quantitative backgrounds and has the goal of educating students on the technical fundamentals of Data Science, with a focus on developing the knowledge and skills needed to transform data into insights. The major is an excellent opportunity for individuals who want to prepare themselves for the exciting Data Scientist positions that are in high demand today.

The Data Science minor is intended for students studying any discipline at Iowa State and is designed to give students an in-depth understanding of data science as it is applied to a variety of domains.

The Data Science certificate is intended for current Iowa State students or anybody who has already earned baccalaureate degree. The program is designed to prepare students for future work with significant Data Science components. The capstone will provide an opportunity for students to apply data science concepts to a domain problem while working in a multi-disciplinary team setting.

Student Learning Outcomes for Data Science Major

After successfully completing the program, students majoring in Data Science will demonstrate:

1. An understanding of and an ability to apply the following data science concepts, tools and methods to data analysis pipelines:
 - a. data acquisition
 - b. data preprocessing
 - c. exploratory data analysis
 - d. inferential and predictive thinking, modeling and analysis
 - e. computational thinking, data structures, and algorithms

2. An understanding of ethical, legal, societal, and economic concerns in the application of data science concepts.
3. An ability to visualize, interpret and communicate the output of data analysis pipelines to stakeholders.
4. An ability to function on multi-disciplinary teams using concepts and tools from data science.

See Undergraduate Minor and Undergraduate Certificate subpages for the respective learning outcomes.

Data Science Major

Purpose

The Bachelor of Science degree program in Data Science is intended for students with strong quantitative backgrounds and has the goal of educating students on the technical fundamentals of data sciences, with a focus on developing the knowledge and skills needed to manage and analyze large-scale, heterogeneous data to address a wide range of problems.

Requirements

The B.S. in Data Science consists of 120 total credit hours including: (1) 39 credit hours in the major core, three credits of which constitute a capstone course that is expected to provide experiential learning; (2) 9 credit hours in an application emphasis area to examine applications and theory of data sciences in a specific area; and (3) 23 credit hours of foundation courses. The capstone course will provide an opportunity for students to apply data science concepts to an application area while working in a multi-disciplinary team setting.

Data Science Major Requirements

Data Science Core Courses

DS 1100	Orientation to Data Science	R
DS 2010	Introduction to Data Science	3
DS 2020	Data Acquisition and Exploratory Data Analysis	3
DS 3030	Concepts and Applications of Machine Learning	3
DS 4010	Data Science Capstone	3
COMS 2280	Introduction to Data Structures	3
COMS 2300	Discrete Computational Structures	3
or CPRE 3100	Theoretical Foundations of Computer Engineering	
COMS 3110	Introduction to the Design and Analysis of Algorithms	3
COMS 3630	Introduction to Database Management Systems	3
CPRE 4190	Software Tools for Large Scale Data Analysis	4
or CPRE 4140	Introduction to Software Systems for Big Data Analytics	
STAT 3201	Intermediate Statistical Concepts and Methods	4
or STAT 3226	Intermediate Business Statistics	

STAT 3447	Probability and Statistical Theory for Data Science	4
STAT 4217	Categorical Data Analysis	3
Total Credits		39

At least 9 credits from the following categories to fulfill the emphasis area:

At least 6 credits from courses at the 3000, 4000, or 5000 level from the following designations:

ABE, ADVRT, AGRON, ANS, ARCH, BCBIO, COMS, CRP, CPRE, CYBE, CYBSC, DS, EE, ECON, FIN, GIS, IE, JLMC, LA, LING, MATH, MIS, MKT, POLS, SE, SOC, STAT, TSM

At least 3 credits from one of the following courses:

BCBIO 4010	Bioinformatics of Sequences	3
BCBIO 4060	Bioinformatics of OMICS	
COMS 3420	Principles of Programming Languages	
COMS 4130	Program Analysis	
MATH 4210	Logic for Mathematics and Computer Science	
COMS 4240	Introduction to High Performance Computing	
COMS 4260	Introduction to Parallel Algorithms and Programming	
COMS 4350	Algorithms for Large Data Sets: Theory and Practice	
COMS 4400	Principles and Practice of Compiling	
COMS 4540	Distributed Systems	
COMS 4740	Introduction to Machine Learning	
CPRE 3880	Mobile Platforms and Applications	
CPRE 4160	Software Evolution and Maintenance	
CPRE 4250	High Performance Computing for Scientific and Engineering Applications	
CPRE 4310	Basics of Information System Security	
EE 4250	Machine learning: A Signal Processing Perspective	
IE 3120	Optimization	
IE 4830	Data Mining	
IE 4870	Big Data Analytics and Optimization	
MATH 3730	Introduction to Scientific Computing	
MATH 4070	Applied Linear Algebra	
MATH 4240	Introduction to High Performance Computing	
MATH 4810	Numerical Methods for Differential Equations	
STAT 4212	Experimental Design and Data Analysis	
STAT 4221	Survey Sampling Design and Data Analysis	
STAT 4207	Multivariate Data Analysis	
Total Credits		9

Toward satisfying pre-requisites and requirements of the College of Liberal Arts and Sciences, the following courses or their equivalents are also required:

LAS 2030	Professional Career Preparation	1
COMS 1270	Introduction to Computer Programming	3
COMS 2270	Object-oriented Programming	4
MATH 1650	Calculus I	4
MATH 1660	Calculus II	4
MATH 2070	Matrices and Linear Algebra	3
MATH 2650	Calculus III	4
STAT 2010	Introduction to Statistical Concepts and Methods	3-4
or STAT 2026	Introduction to Business Statistics	
World Language 3 years in high school or 1 year in college		0 - 8
Natural Science		8
Social Science		9
Arts and Humanities		12

The following courses meet the communication proficiency requirement:

LIB 1600	Introduction to College Level Research	1
ENGL 1500	Critical Thinking and Communication	3
ENGL 2500	Written, Oral, Visual, and Electronic Composition	3
One of the following:		
ENGL 3020	Business Communication	3
ENGL 3140	Technical Communication	3

As majors in the College of Liberal Arts and Sciences, Data Science students must meet College of Liberal Arts and Sciences (<https://catalog.iastate.edu/collegeofliberalartsandsciences/#lascollegerequirementstext>) and University-wide requirements (<https://catalog.iastate.edu/collegescurricula/>) for graduation in addition to those stated above for the major.

LAS majors require a minimum of 120 credits, including a minimum of 45 credits at the 3000/4000 level or above taken at a four-year college. All such credits, including courses taken on a pass/not pass basis, may be used to meet this requirement. You must also complete the LAS world language requirement and career proficiency requirement.

Students in all ISU majors must complete a three-credit course in U.S. Cultures and Communities and a three-credit course in International Perspectives. Check (<http://www.registrar.iastate.edu/courses/div-ip-guide.html>) for a list of approved courses. Discuss with your advisor how the two courses that you select can be applied to your graduation plan.

According to the university-wide Communication Proficiency Grade Requirement (<https://catalog.iastate.edu/academics/>)

#communicationproficiencypolicytext), students must demonstrate their communication proficiency by earning a grade of C or better in ENGL 2500. The Data Science program requires a C or higher in the upper-level ENGL course (3020, 3140, or 3320).

Four Year Plan

B.S., Data Science

Freshman

Fall	Credits Spring	Credits
DS 1100	R MATH 1660	4
MATH 1650	4 COMS 2270	4
COMS 1270	3 DS 2020	3
ENGL 1500	3 Social Science	3
LIB 1600	1	
DS 2010	3	
	14	14

Sophomore

Fall	Credits Spring	Credits
MATH 2650	4 MATH 2070	3
COMS 2280	3 COMS 2300 or CPRE 3100	3
Arts and Humanities/U.S. Cultures and Communities	3 Social Science/International Perspectives	3
STAT 2010	4 Arts and Humanities	3
ENGL 2500	3 LAS 2030	1
	STAT 3201	4
	17	17

Junior

Fall	Credits Spring	Credits
DS 3030	3 COMS 3630	3
COMS 3110	3 STAT 4217	3
STAT 3447	4 Arts and Humanities	3
Arts and Humanities (3000+ level)	3 Natural Science	4
Elective or World Language	3-4 Elective or World Language	3-4
	16-17	16-17

Senior

Fall	Credits Spring	Credits
Application Emphasis Area	3 DS 4010	3
Application Emphasis Area	3 CPRE 4190	4
ENGL 3020 or 3140	3 Application Emphasis Area	3
Natural Science	4 Social Science (3000+ Level)	3
	13	13

Total Credits: 120-122

The major elective courses will come from Data Science emphasis areas as outlined on the Undergraduate Major page. This includes 3 credits from a list of elective courses and an additional 6 credits at the 3000-level or above from a wide variety of designations.

All students are required to take at least 45 hours of courses at the 3000-level or above. This may require taking additional electives.

Data Science Minor

Purpose

The minor in data science is intended for students studying any discipline at Iowa State and is designed to give students an in-depth understanding of data science as it is applied to a variety of domains. The minor in data science will prepare students with the technical and communication skills to enter the workforce as domain experts with data science skills.

Learning Outcomes for Data Science Minor

After completing the minor in data science, students will demonstrate:

- An ability to apply data science concepts, tools and technologies to data analysis pipelines,
- An understanding of ethical, legal, societal, and economic concerns in application of data science concepts,
- An ability to visualize, interpret and communicate the output of data analysis pipelines to stakeholders, and
- An ability to function on multi-disciplinary teams using concepts and tools from data science.

Requirements

The minor in data science requires the completion of 15 credit hours, including 9 credits from the data science core and 6 credits from approved data science electives.

At least 6 credits in courses numbered 3000-level with a grade of C or higher.

At least 3 credits taken at Iowa State.

At least 3 credits that are not used to meet any other department, college, or university requirement.

Courses for the minor cannot be taken on a pass/not-pass basis.

Course Requirements for Data Science Minor

Core Courses (9 credits)

DS 2010	Introduction to Data Science (Required)	3
DS 2020	Data Acquisition and Exploratory Data Analysis (Required)	3
DS 3010	Applied Data Modeling and Predictive Analysis (Required)	3

* DS 3010 has a prerequisite of an introductory statistics course:
STAT 1010, STAT 1040, STAT 2010, STAT 2026, STAT 3031,
STAT 3005, STAT 3022, or STAT 3030.

Electives (6 credits)		
ABE 3160	Applied Numerical Methods for Agricultural and Biosystems Engineering	3
ADVRT 3350	Advertising Media Planning	3
ADVRT 4970	Special Topics in Communication	1-3
AGRON 2700	Geospatial Technologies	3
AGRON 4250	Crop and Soil Modeling	3
BCBIO 3220	Introduction to Bioinformatics and Computational Biology	3
COMS 3110	Introduction to the Design and Analysis of Algorithms	3
COMS 3630	Introduction to Database Management Systems	3
COMS 4240	Introduction to High Performance Computing	3
COMS 4350	Algorithms for Large Data Sets: Theory and Practice	3
COMS 4530	Privacy Preserving Algorithms and Data Security	3
COMS 4740	Introduction to Machine Learning	3
CRP 2510	Fundamentals of Geographic Information Systems	3
CRP 3510	Geospatial Analysis and Visualization	3
CRP 4520	Geographic Data Management and Planning Analysis	3
CRP 4540	Fundamentals of Remote Sensing and Spatial Analysis	3
CRP 4560	GIS Programming and Automation	3
CPRE 4190	Software Tools for Large Scale Data Analysis	4
CPRE 4260	Introduction to Parallel Algorithms and Programming	4
ECON 3710	Introductory Econometrics	4
EE 4280	Image Analysis using Machine Learning	3
FIN 4500	Analytical Methods in Finance	3
IE 3120	Optimization	3
IE 4830	Data Mining	3
LA 5580	Web Mapping and Spatial Data Visualization	3
LING 4100	Language as Data	3
MATH 3040	Combinatorics	3
MATH 3140	Graph Theory	3
MATH 3730	Introduction to Scientific Computing	3
MATH 4220X	Mathematical Principles of Data Science	3
MIS 4360	Introduction to Business Analytics	3
MIS 4460	Advanced Business Analytics	3

MKT 4680	Analytics for Marketing Strategies	3
STAT 3201	Intermediate Statistical Concepts and Methods	4
STAT 3030	Probability and Statistics for Computer Science	3
STAT 4279	Statistical Computing for Data Analysis	3
STAT 4207	Multivariate Data Analysis	3
STAT 4217	Categorical Data Analysis	3
STAT 4232	Empirical Methods for the Computational Sciences	3
TSM 4330	Precision Agriculture	3

The Data Science Undergraduate Minor is an LAS Minor. In addition to University policies governing minors (<https://catalog.iastate.edu/academics/#degreeplanningtext>), LAS minors require at least 6 credits in courses numbered 3000-level or above, with a grade of C or higher.

Data Science Certificate

Purpose

The certificate in data science is intended for students studying any discipline at Iowa State and is designed to prepare them for future work with significant data science components. The data science certificate is also available to students who have already earned a Baccalaureate degree from Iowa State or elsewhere. The capstone will provide an opportunity for students to apply data science concepts to a domain problem while working in a multi-disciplinary team setting. The certificate in data science will prepare students with the technical and communication skills to enter the workforce as domain experts with data science skills.

Learning Outcomes for Data Science Certificate

After completing the certificate in data science, students will demonstrate:

- An ability to apply data science concepts, tools and technologies to data analysis pipelines,
- An understanding of ethical, legal, societal, and economic concerns in application of data science concepts,
- An ability to visualize, interpret and communicate the output of data analysis pipelines to stakeholders, and
- An ability to function on multi-disciplinary teams using concepts and tools from data science.

Requirements

The certificate in data science requires the completion of 21 credit hours, including 9 credits from the data science core, 9 credits from approved data science electives, and a three-credit data science capstone experience.

At least 9 credits must be taken in courses numbered at the 3000-level or above. At least 6 credits must be taken at Iowa State University.

Courses for the certificate cannot be taken on a pass/not-pass basis.

Course Requirements for Data Science Certificate

Core Courses (9 credits)

DS 2010	Introduction to Data Science (Required)	3
DS 2020	Data Acquisition and Exploratory Data Analysis (Required)	3
DS 3010	Applied Data Modeling and Predictive Analysis (Required)	3

* DS 3010 has a prerequisite of an introductory statistics course:

STAT 1010, STAT 1040, STAT 2010, STAT 2026, STAT 3031, STAT 3005, STAT 3022, or STAT 3030.

Electives (9 credits)

ABE 3160	Applied Numerical Methods for Agricultural and Biosystems Engineering	3
ADVRT 3350	Advertising Media Planning	3
ADVRT 4970	Special Topics in Communication	1-3
BCBIO 3220	Introduction to Bioinformatics and Computational Biology	3
COMS 3110	Introduction to the Design and Analysis of Algorithms	3
COMS 3630	Introduction to Database Management Systems	3
COMS 4240	Introduction to High Performance Computing	3
COMS 4350	Algorithms for Large Data Sets: Theory and Practice	3
COMS 4530	Privacy Preserving Algorithms and Data Security	3
COMS 4740	Introduction to Machine Learning	3
CRP 2510	Fundamentals of Geographic Information Systems	3
CRP 3510	Geospatial Analysis and Visualization	3
CRP 4520	Geographic Data Management and Planning Analysis	3
CRP 4560	GIS Programming and Automation	3
CPRE 4190	Software Tools for Large Scale Data Analysis	4
CPRE 4260	Introduction to Parallel Algorithms and Programming	4
ECON 3710	Introductory Econometrics	4
FIN 4500	Analytical Methods in Finance	3
IE 3120	Optimization	3
IE 4830	Data Mining	3
LING 4100	Language as Data	3
MATH 3040	Combinatorics	3
MATH 3140	Graph Theory	3
MATH 3730	Introduction to Scientific Computing (MATH 4220X::Mathematical Principles of Data Science)	3
MATH 4220X	Mathematical Principles of Data Science	3

MIS 4360	Introduction to Business Analytics (::Mathematical Principals of Data Science)	3
MIS 4460	Advanced Business Analytics	3
MKT 4680	Analytics for Marketing Strategies	3
STAT 3201	Intermediate Statistical Concepts and Methods	4
STAT 3030	Probability and Statistics for Computer Science	3
STAT 4279	Statistical Computing for Data Analysis	3
STAT 4207	Multivariate Data Analysis	3
STAT 4217	Categorical Data Analysis	3
STAT 4232	Empirical Methods for the Computational Sciences	3
Data Science capstone experience (3 credits)		
DS 4010	Data Science Capstone	3

The Data Science Certificate is an LAS Certificate and must meet University-wide policies for undergraduate certificates (<https://catalog.iastate.edu/academics/#degreeplanningtext>).